

Weapon-Target Assignment Model Improvements on

Multi-Target Interception and Attack-Oriented WTA Model
Comparative Optimization with ABC, PSO, and Genetic Algorithm

3

ALGORITHM

10

MISSILE

5

DECOY

10K

ITERATIONS

6

CONSTRAINT TYPE

Contents

- **Definition of the Problem**
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- **Results of the Attack Model**
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- **Future Works**

Definition of the Problem

What is Weapon-Target Assignment (WTA)?

- **Weapon-Target Assignment (WTA):** The optimal assignment of available missiles to threats.
- **Original Scenario (Defense):** Stopping incoming missiles with our own missiles.
- **NP-Hard Complexity:** As the number of missiles and targets increase, solution space grows exponentially.
- **Kinematic Approach:** Assignment is not just numerical; it's done by calculating the positions and vectors of targets in the x-y space.

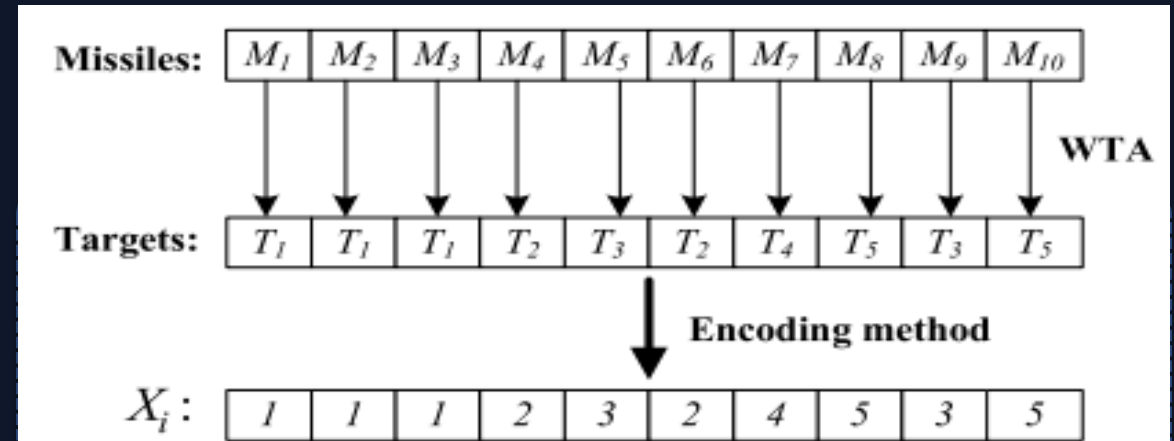
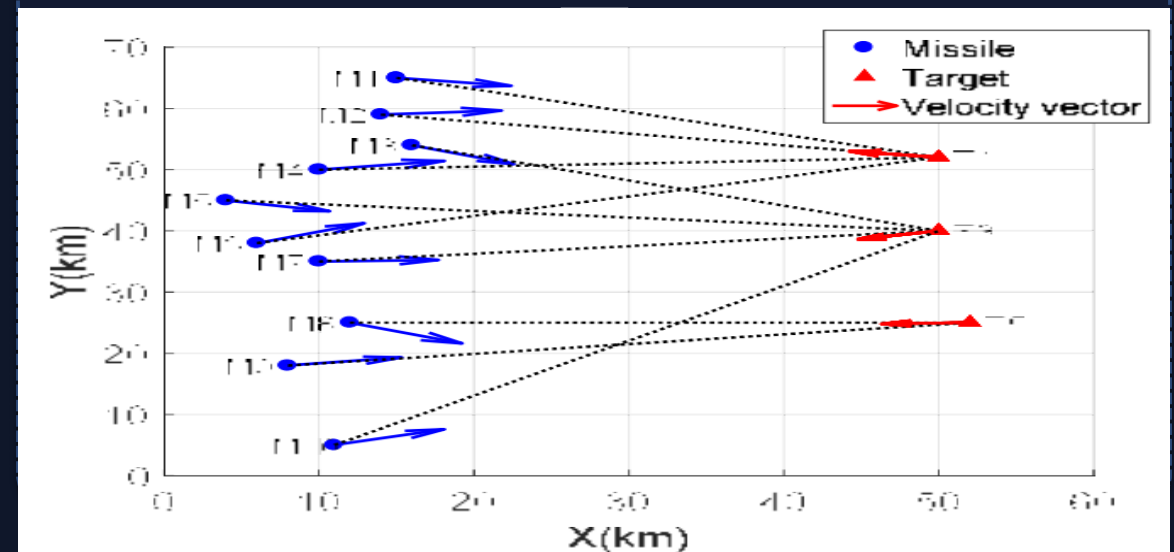


FIGURE 3. Illustration of encoding method.



Why is it NP-Hard?

NP-Hard Structure and Combinatorial Explosion

- Number of possible assignments for 10 missiles and 5 targets: $5^{10} = 9,765,625$ different combinations.
- When Decoys are added (15-unit vector): $6^{15} \approx 470$ billion combinations.
- Traditional methods fail at this scale.
- Therefore, Metaheuristic Algorithms are used.
- They do not guarantee the optimal solution, but they find pretty good solutions within acceptable time.

Scenario	Variable	Solution Space
Old Model (Missiles Only)	10	~10 million
New Model (Missile+Decoy)	15	~470 billion
Large Scenario (50 units)	50	$\sim 10^{38}$
Comparison: Atoms in the universe	—	$\sim 10^{80}$

Critical Guidance Parameters: Miss Distance, Time-to-Go, and LOS

1. Miss Distance (ΔS):

Meaning: Closest distance to the target during flight.

Why Minimize? Closer to zero = Maximum lethality.

2. Time-to-Go (tgo):

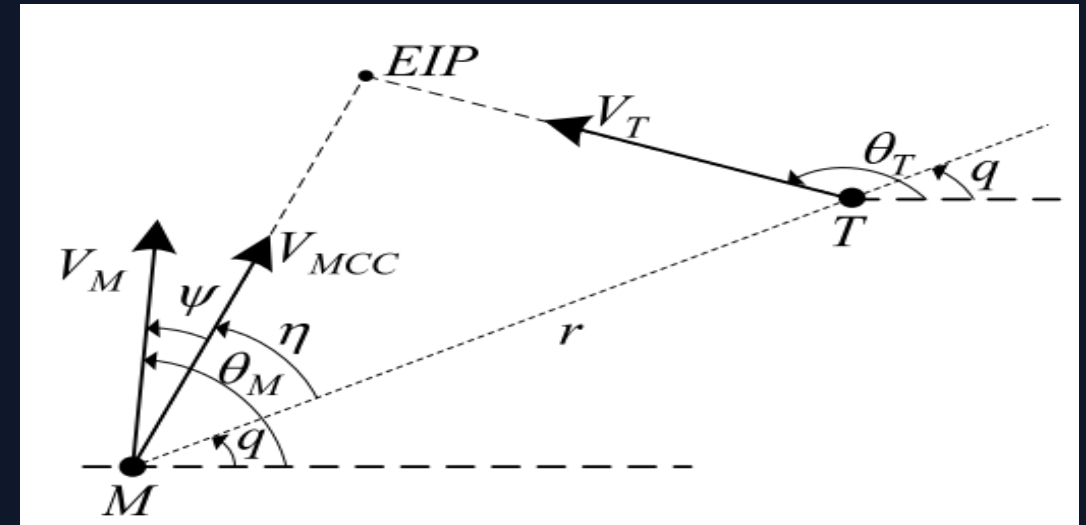
Meaning: Estimated time to hit the target

Why Minimize? Gives the target zero time to evade or jam.

3. LOS (Line of Sight):

Meaning: Direct line connecting missile and target. Its change rate is LOS Rate (λ').

Why Minimize? Keep LOS rate at zero. No change in LOS angle = Definite collision course.



Approaches in the Literature

Traditional Methods

Lagrange Relaxation, Exhaustive Search, Branch & Bound. Accurate at small scale, but fails on large problems.

Genetic Algorithms (GA)

Inspired by natural selection. Explores the search space via Selection, Crossover, and Mutation. It is compatible with integer problems.

Artificial Bee Colony (ABC)

Simulates the foraging behavior of honey bees. The main algorithm proposed by the paper. It converges to the optimum value, thanks to employee, onlooker and scout bees.

Particle Swarm Optimization (PSO)

Models the collective movement of bird flocks. Works in continuous space, adapted to integer using floor(). It converges to the optimum value, thanks to speed and location updates.

Solution Approach Proposed by the Paper

- **Kinematic Phit Calculation:** Miss Distance, Time-to-Go, LOS(Line of Sight) are considered to calculate Phit value.
- **Grouping Constraints:** Prevents excessive accumulation of ammunition on certain targets.
- **Penalty Function:** Penalties are added to the cost function for solutions that violate constraints.
- **Objective Function:** Algorithms minimize the expected survival value of targets.

Feature	Fixed	Adaptive
Assignment Rule	Exact number	Min-Max range
Flexibility	Low	High
Example (Target 1)	Exactly 4 missiles	1–4 missiles
Penalty Formula	$Q_j \cdot G_j$	$U_j \cdot E_j + L_j \cdot F_j$
Solution Space	Narrow	Wide

$$\min \left(\sum_{i=1}^m \left\{ V_j \prod_{i=1}^m (1 - P_{ij})^{X_{ij}} + Q_j G_j \right\} + \sum_{i=1}^m S_i H_i \right) \quad (15)$$

For Fixed Grouping

$$\min \left(\sum_{j=1}^n \left\{ V_j \prod_{i=1}^m (1 - P_{ij})^{X_{ij}} + U_j E_j + L_j F_j \right\} + \sum_{i=1}^m S_i H_i \right) \quad (18)$$

For Adaptive Grouping

How is Kinematic Hit Probability (P_{ij}) Calculated?

-> Paper's Probability Function (Equations 8, 9, and 10)

- Miss Distance:
- Time-to-Go:
- Line of Sight:

$$\begin{cases} P_{\Delta S(i,j)} = P_0^{\Delta S} e^{-\frac{1}{2} \left(\frac{\Delta S(i,j)}{\delta_{\Delta S}} \right)^2} \\ P_{T_{go}(i,j)} = P_0^{T_{go}} e^{-\frac{1}{2} \left(\frac{T_{go}(i,j)}{\delta_{T_{go}}} \right)^2} \\ P_{\dot{q}(i,j)} = P_0^{\dot{q}} e^{-\frac{1}{2} \left(\frac{\dot{q}(i,j)}{\delta_{\dot{q}}} \right)^2} \end{cases} \quad (8)$$

- P_{ij} (probability of missile i hitting target j) is modeled by AI as a negative exponential function of three kinematic indicators (Eq. 8).

$$\begin{cases} \delta_{\Delta S} = \frac{\sum_{j=1}^n \sum_{i=1}^m \Delta S(i,j)}{mn} \\ \delta_{T_{go}} = \frac{\sum_{j=1}^n \sum_{i=1}^m T_{go}(i,j)}{mn} \\ \delta_{\dot{q}} = \frac{\sum_{j=1}^n \sum_{i=1}^m \dot{q}(i,j)}{mn} \end{cases} \quad (9)$$

- The average value of these 3 parameters is calculated separately and written as the denominator in the exponent in Equation (8):

$$P_{ij} = \beta_{\Delta S} P_{\Delta S(i,j)} + \beta_{T_{go}} P_{T_{go}(i,j)} + \beta_{\dot{q}} P_{\dot{q}(i,j)} \quad (10)$$

- The paper combines these three indicators with balancing weights (β) to find the final hit probability:

Working Mechanisms of the Algorithms

Iteration steps and basic formulas of the meta-heuristic algorithms used

Artificial Bee Colony (ABC)

- 1. Employed Bee Phase
- 2. Onlooker Bee Phase
- 3. Scout Bee Phase

Position Update:

$$v_{ij} = x_{ij} + \phi(x_{ij} - x_{kj})$$

Particle Swarm Optimization (PSO)

- 1. Velocity Update
- 2. Position Update
- 3. Transformation

Velocity / Position Formulas:

$$\begin{aligned} v_i &= w \cdot v_i + c1 \cdot r1(pbest) \\ &\quad + c2 \cdot r2(gbest) \\ x_i &= x_i + v_i \end{aligned}$$

Genetic Algorithm (GA)

- 1. Selection (Tournament)
- 2. Crossover
- 3. Mutation

Crossover Example (p_c=0.8):

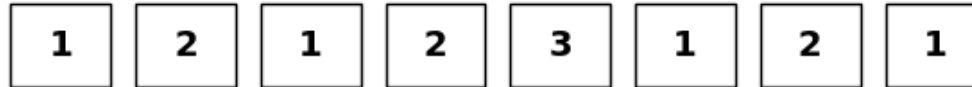
```
P1: [1 1 | 2 3]
P2: [0 4 | 5 5]
Child: [1 1 | 5 5]
```

ABC Algorithm: 2-Bee Execution Example

ABC Algorithm: Initial State

Initialization: A population of N arrays is generated. (Showing 2 for simplicity)

Food Source 1
(Bee 1)



Cost: 2.45

Food Source 2
(Bee 2)



Cost: 3.10

PSO Algorithm: Discrete WTA Execution Example

PSO Algorithm: Initialization

Particle has Velocity (V), Continuous Position (P), and Discrete Targets.

Velocity (V)

0.5	-0.2	0.8	-0.6	1.1	-0.3	0.4	-0.9
-----	------	-----	------	-----	------	-----	------

Position (P)
Continuous

1.2	2.8	1.1	2.4	3.9	1.5	2.1	1.3
-----	-----	-----	-----	-----	-----	-----	-----

Position (P)
Discrete

1	2	1	2	3	1	2	1
---	---	---	---	---	---	---	---

Cost: 2.45

gBest
Discrete

3	3	2	1	2	1	1	3
---	---	---	---	---	---	---	---

Cost: 1.90

Genetic Algorithm: Discrete WTA Execution Example

Genetic Algorithm: Tournament Selection

Two Parent arrays are selected based on lowest Cost.

Parent 1

Cost: 2.45

Parent 2

3	3	2	1	2	1	1	3
---	---	---	---	---	---	---	---

Cost: 3.10

Offspring

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Meaning of Parameters and Constant Values to Apply ABC Algorithm

TABLE 3. Parameter setting for the WTA method.

Parameters	Meaning	Values
N	Navigation constant	4
τ	Time constant of flight control system	1
$P_0^{\Delta S}$	Default interception probability for miss-distance	0.9
$P_0^{T_{go}}$	Default interception probability for time-to-go	0.9
P_0^q	Default interception probability for LOS rate	0.9
$\beta_{\Delta S}$	Weight of intercepting probability of Miss-distance	0.5
$\beta_{T_{go}}$	Weight of intercepting probability of time-to-go	0.25
β_q	Weight of intercepting probability of LOS rate	0.25
Q_j	Penalty factor for jth target	0.1
S_i	Penalty factor for ith missile	0.04
U_j	Upper bound penalty factor for jth target	0.1
L_j	Lower bound penalty factor for jth target	0.1

TABLE 4. Parameter setting for the ABC algorithm.

Parameters	Meaning	Values
D	Dimension of the solutions	M
S	Initial population of the employed bees and onlooker bees	100
$limit$	Repeated restriction times of algorithm	100
TC	Termination condition of ABC algorithm	$1000 * D$

Assignment Results — Case 1 (3 Targets)

TABLE 1. Parameters of the 10 missiles.

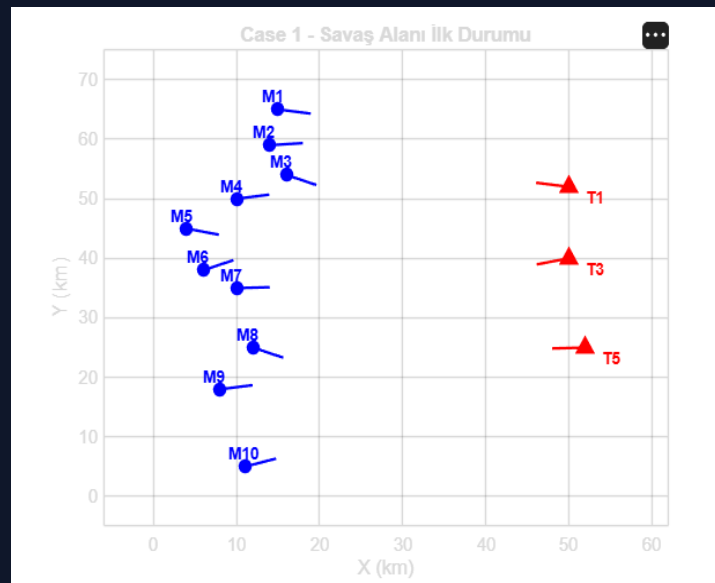
Missile	X (km)	Y (km)	Heading angle (deg)	Velocity (m/s)
M1	15	65	-10	750
M2	14	59	5	780
M3	16	54	-25	725
M4	10	50	10	820
M5	4	45	-15	690
M6	6	38	25	765
M7	10	35	2	770
M8	12	25	-25	795
M9	8	18	10	745
M10	11	5	20	755

TABLE 2. Parameters of the 5 targets.

Target	X (km)	Y (km)	Heading angle (deg)	Velocity (m/s)	Value
T1	50	52	170	580	0.85
T2	48	45	175	560	0.69
T3	50	40	195	535	0.72
T4	46	34	170	550	0.95
T5	52	25	182	565	0.80

Fixed Constraint

T1 -> 4 | T3 -> 4 | T5 -> 2

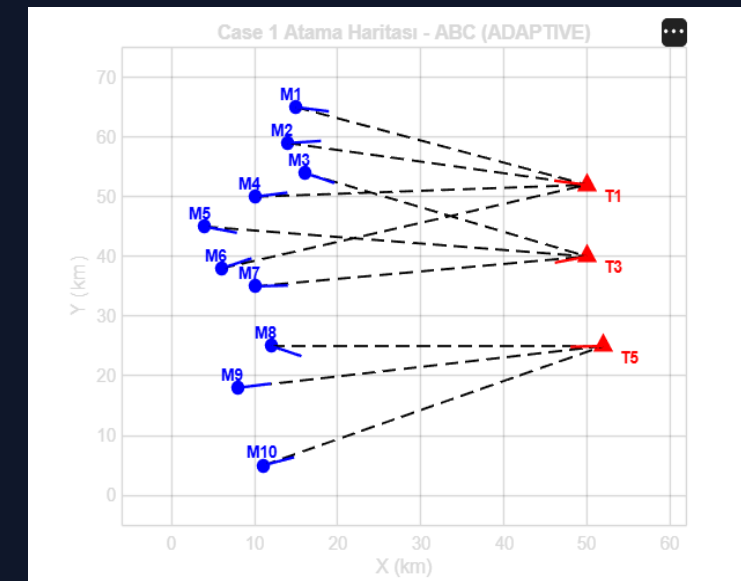
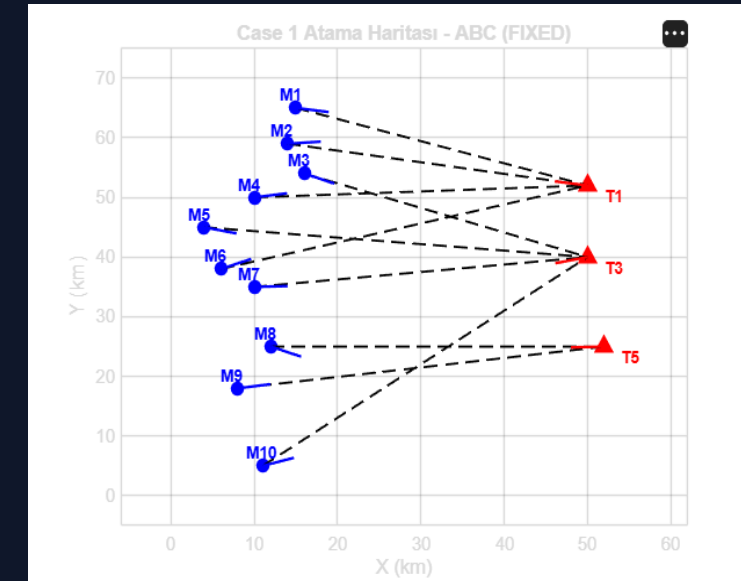


Adaptive Constraint

Min: 1

Max: 4

For each target



Paper Model (Defense) Verification Case1

The model proposed in the paper was rewritten and verified by me, and GA was added. There are 10 missiles and 3 targets in Case 1.

CASE 1 (3 Targets) - FIXED

Algorithm	Cost	Paper Result
ABC	0.1252	0.0525

CASE 1 (3 Targets) - ADAPTIVE

Algorithm	Cost	Paper Result
ABC	0.0790	0.0241

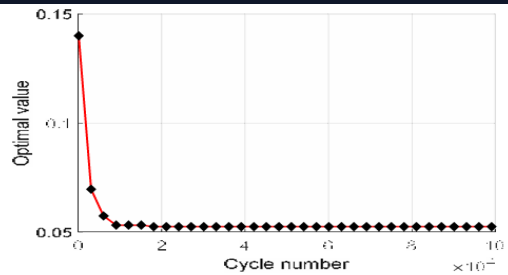


FIGURE 6. Objective function for Case 1 with fixed grouping strategy.

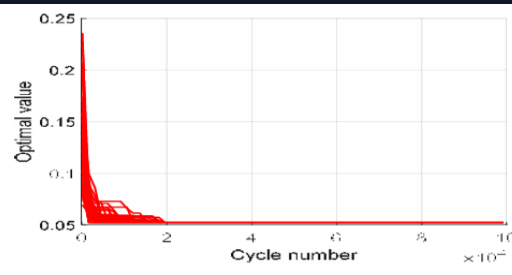


FIGURE 7. Objective function for Case 1 with fixed grouping strategy (50 runs).

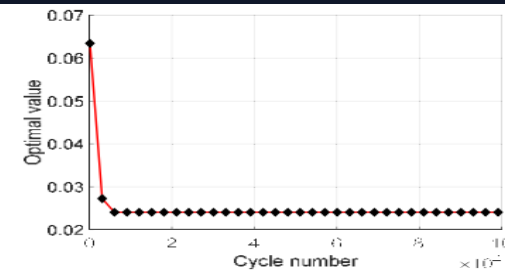


FIGURE 9. Objective function for Case 1 with adaptive grouping strategy.

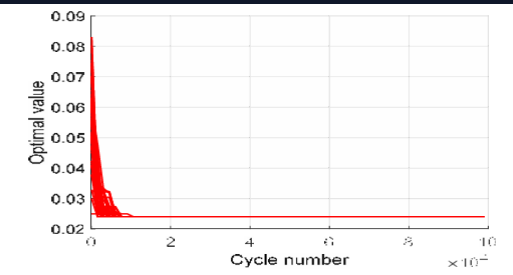
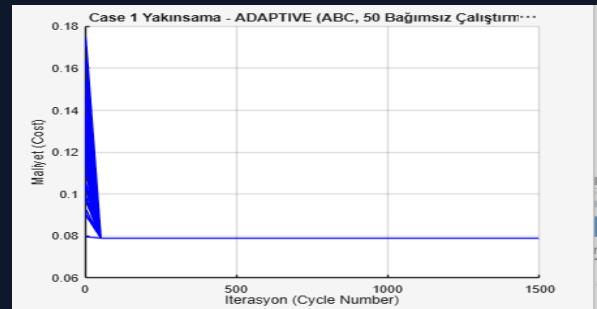
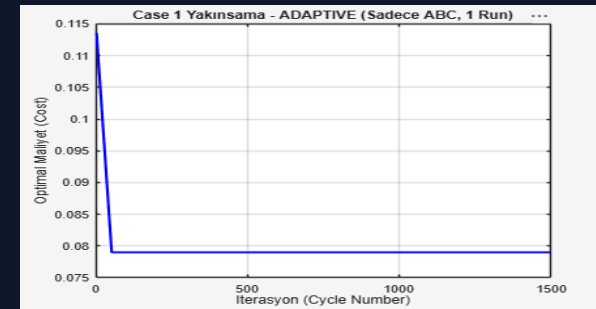
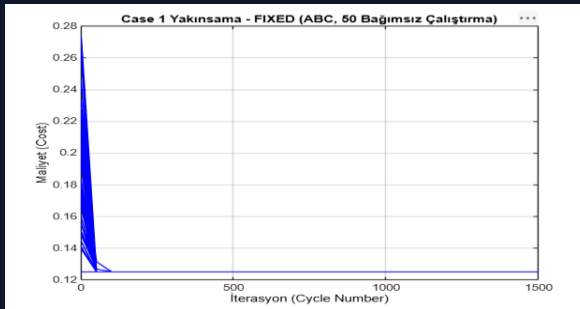
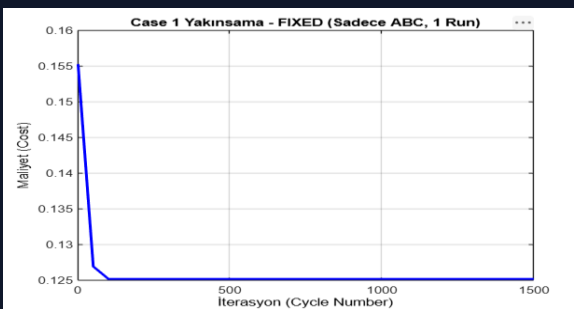


FIGURE 10. Objective function for Case 1 with adaptive grouping strategy (50 runs).



Assignment Results — Case 2 (5 Targets)

TABLE 1. Parameters of the 10 missiles.

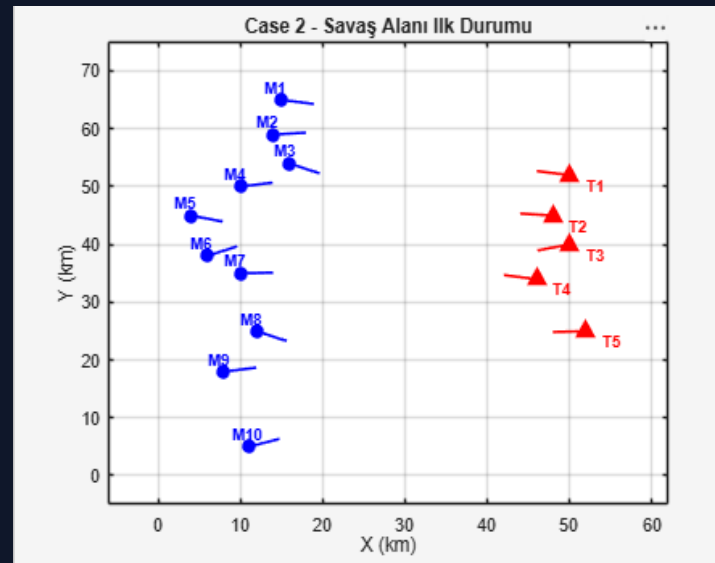
Missile	X (km)	Y (km)	Heading angle (deg)	Velocity (m/s)
M1	15	65	-10	750
M2	14	59	5	780
M3	16	54	-25	725
M4	10	50	10	820
M5	4	45	-15	690
M6	6	38	25	765
M7	10	35	2	770
M8	12	25	-25	795
M9	8	18	10	745
M10	11	5	20	755

TABLE 2. Parameters of the 5 targets.

Target	X (km)	Y (km)	Heading angle (deg)	Velocity (m/s)	Value
T1	50	52	170	580	0.85
T2	48	45	175	560	0.69
T3	50	40	195	535	0.72
T4	46	34	170	550	0.95
T5	52	25	182	565	0.80

Fixed Constraint

T1 -> 2 | T2 -> 1 | T3 -> 2 | T4 -> 3 | T5 -> 2

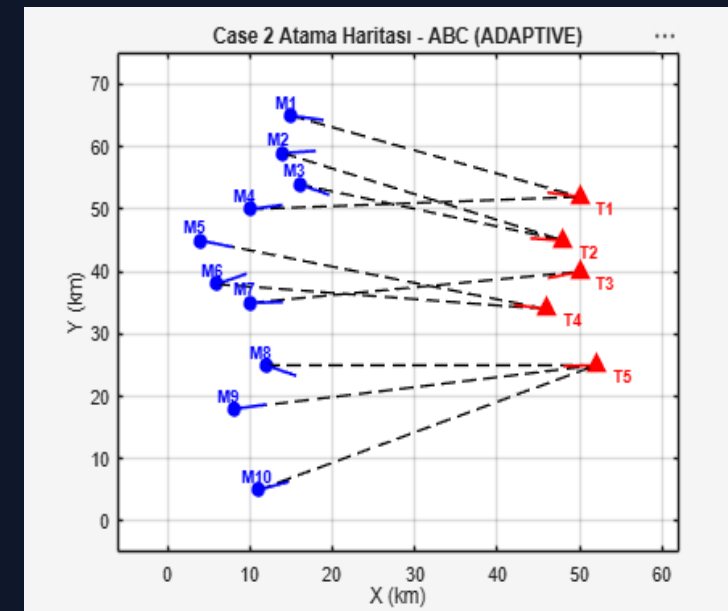
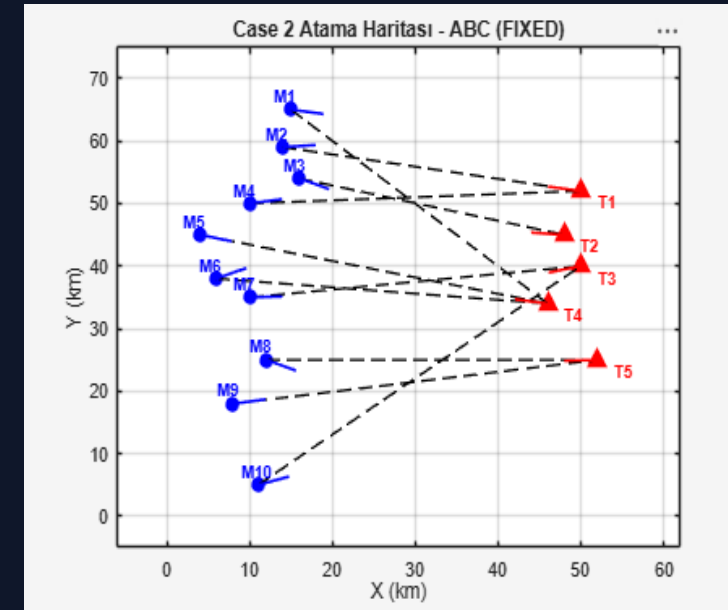


Adaptive Constraint

Min: 1

Max: 3

For each target



Paper Model (Defense) Verification Case 2

The model proposed in the paper was rewritten and verified by me, and GA was added. There are 10 missiles and 5 targets in Case 2.

CASE 2 (5 Targets) - FIXED

Algorithm	Cost	Paper Result
ABC	0.5939	0.3153

CASE 2 (5 Targets) - ADAPTIVE

Algorithm	Cost	Paper Result
ABC	0.5407	0.2672

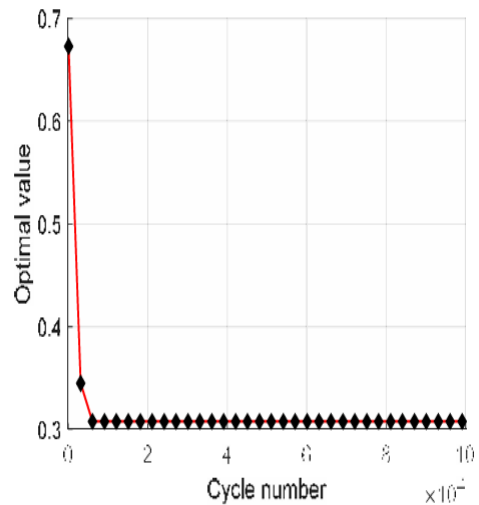


FIGURE 12. Objective function for Case 2 with fixed grouping strategy.

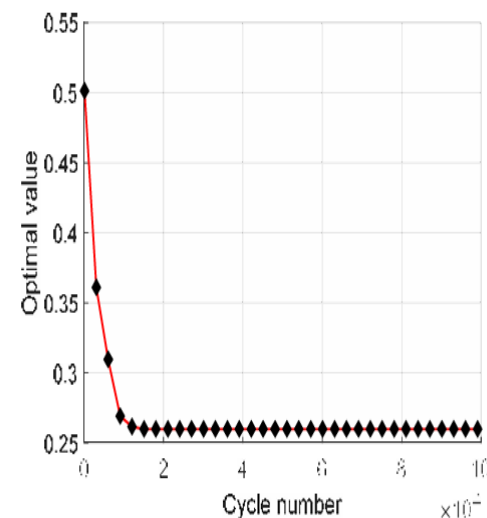
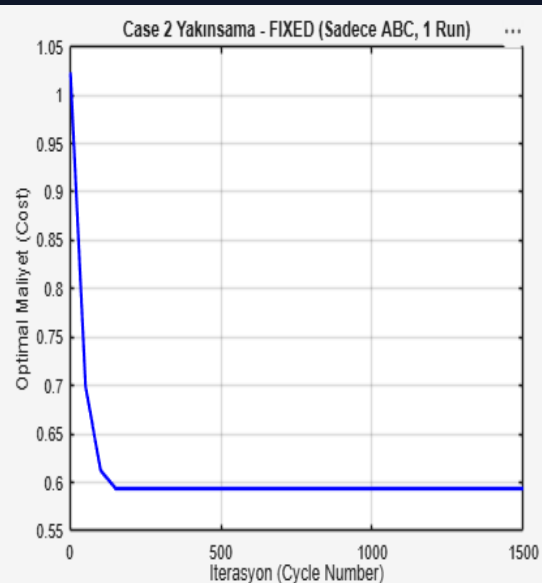
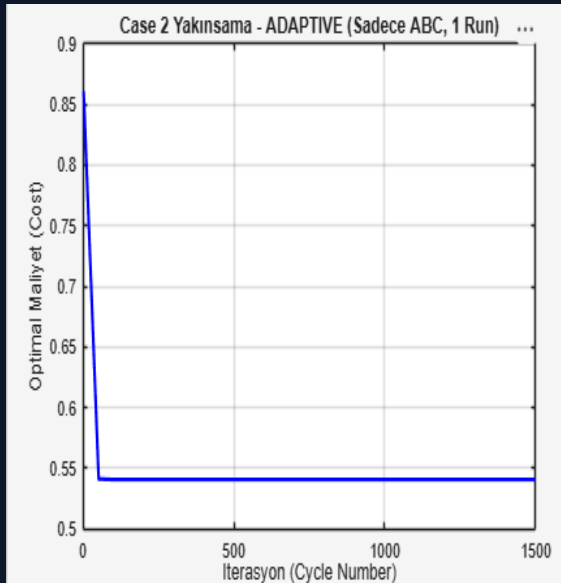


FIGURE 14. Objective function for Case 2 with adaptive grouping strategy.



Case 2 50 Iterations ABC and PSO Comparison

TABLE 7. Comparison of ABC and PSO algorithm for Case 2.

Index	Adaptive Grouping		Fixed Grouping	
	ABC	PSO	ABC	PSO
Number of convergence cases	50	33	50	31
Average convergence time (s)	0.29	1.30	0.31	1.43

Normal(Standard) PSO

Index	Adaptive ABC	Adaptive PSO	Fixed ABC	Fixed PSO
Number of convergence cases	50	19	50	6
Average convergence time(s)	0.05	0.37	0.27	0.01

Hybrid PSO

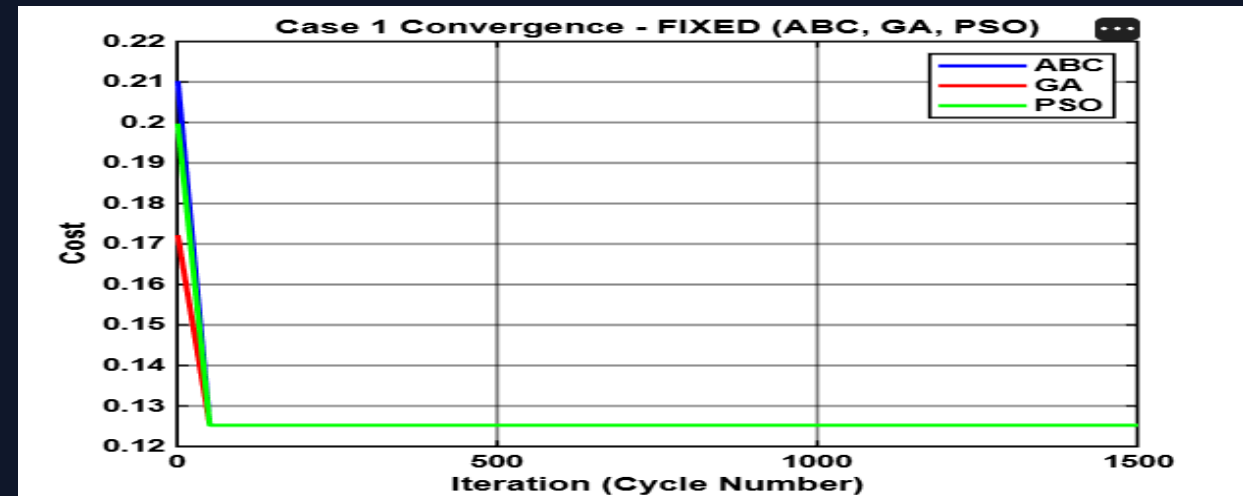
Index	Adaptive ABC	Adaptive PSO	Fixed ABC	Fixed PSO
Number of convergence cases	50	50	50	34
Average convergence time(s)	0.05	0.06	0.31	0.61

Additional Comparisons for Case 1 (3 targets)

ABC, PSO and also Genetic Algorithm were used to compare whether they reach the optimum cost value.

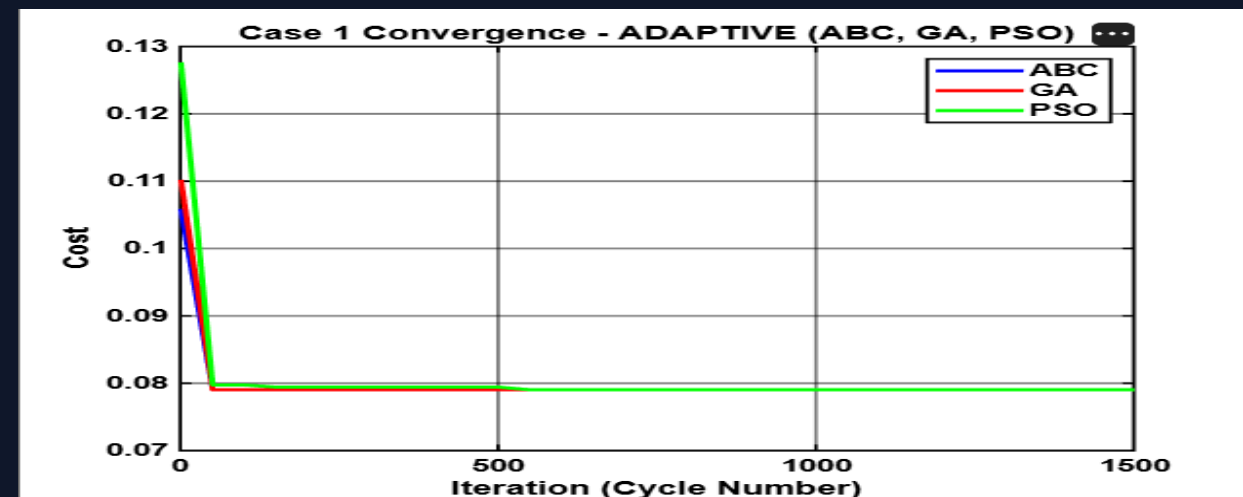
CASE 1 (3 Targets) - FIXED

Algorithm	Cost	Status
ABC	0.1252	OPTIMUM ✓
GA	0.1252	OPTIMUM ✓
PSO	0.1252	OPTIMUM ✓



CASE 1 (3 Targets) - ADAPTIVE

Algorithm	Cost	Status
ABC	0.0790	OPTIMUM ✓
GA	0.0790	OPTIMUM ✓
PSO	0.0790	OPTIMUM ✓

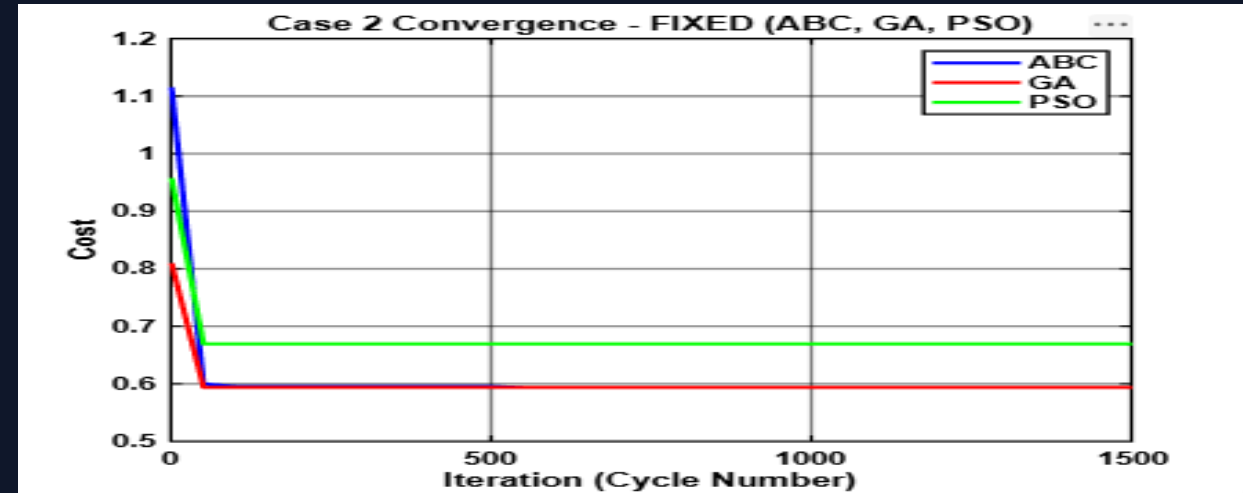


Additional Comparisons for Case 2 (5 targets)

ABC, PSO and also Genetic Algorithm were used to compare whether they reach the optimum cost value.

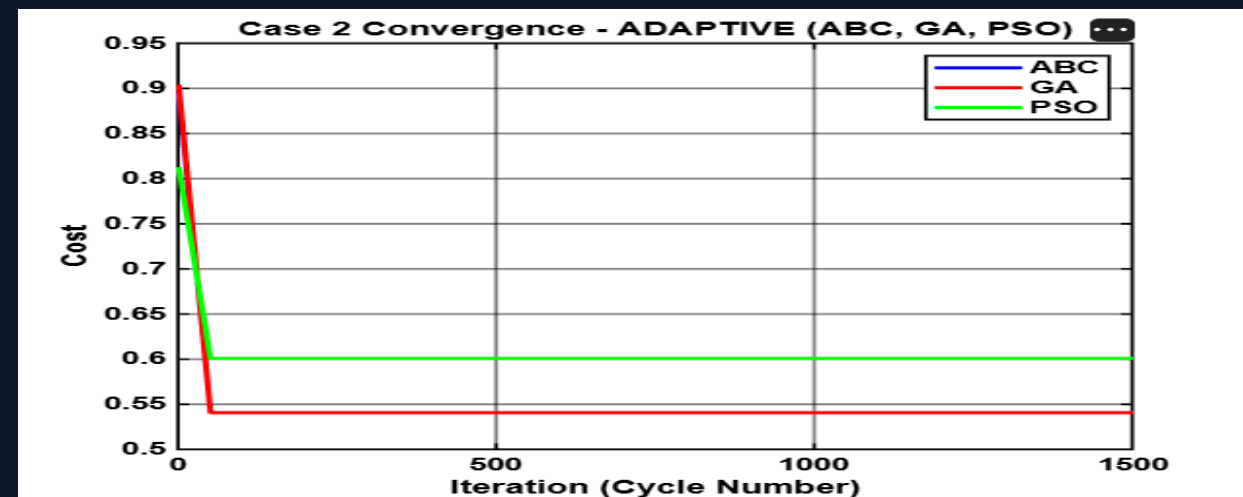
CASE 2 (5 Targets) - FIXED

Algorithm	Cost	Status
ABC	0.5939	OPTIMUM ✓
GA	0.5939	OPTIMUM ✓
PSO	0.6643	Local Min ⚠



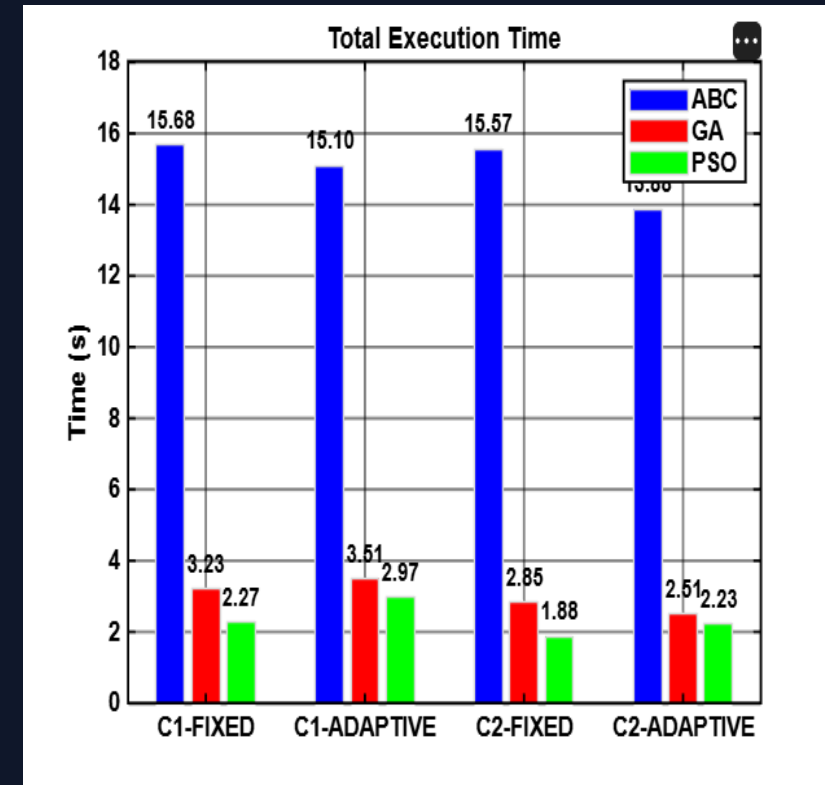
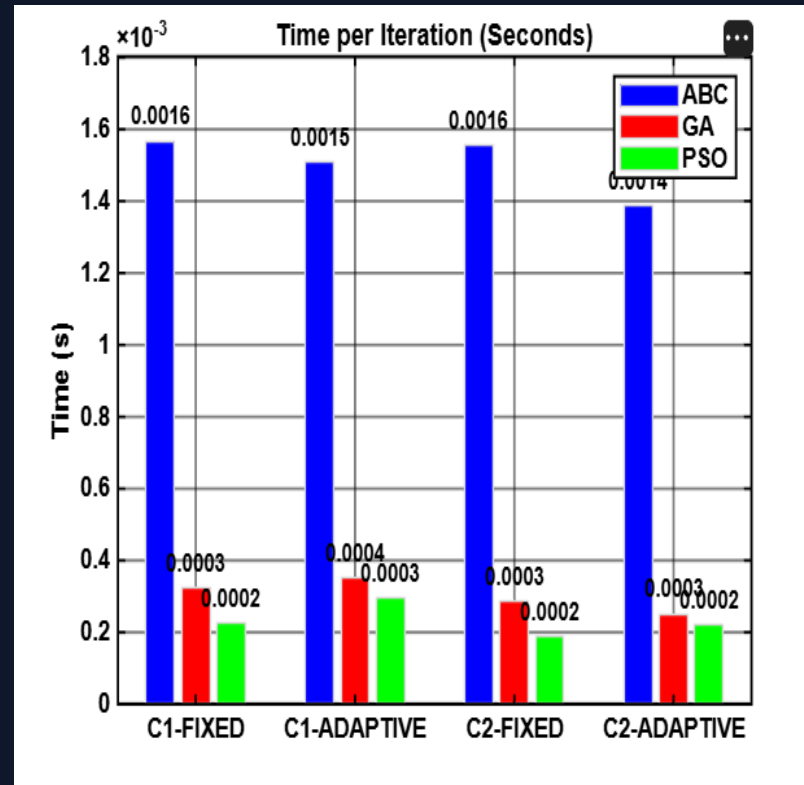
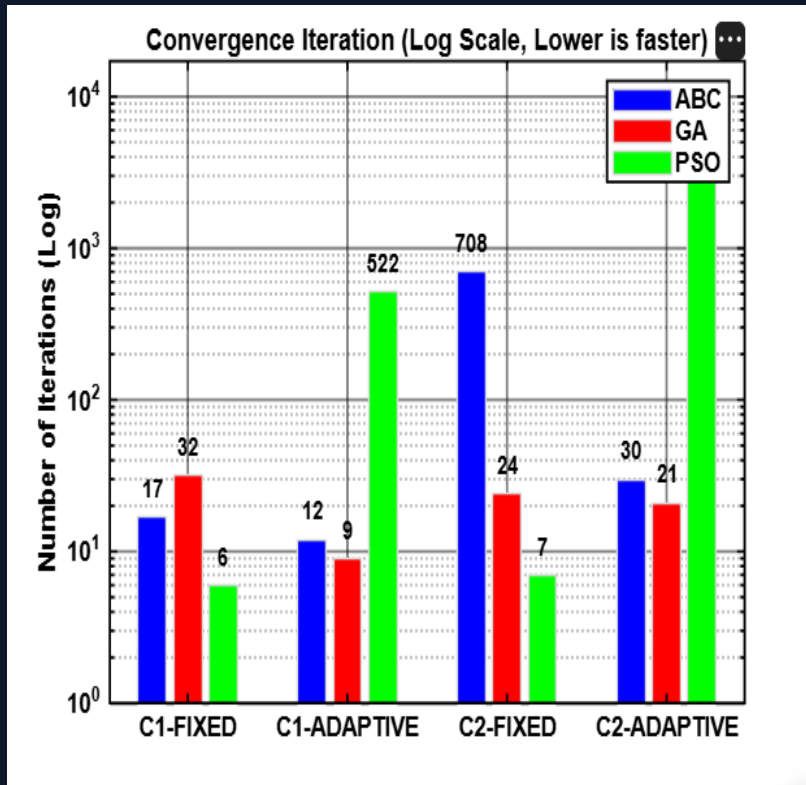
CASE 2 (5 Targets) - ADAPTIVE

Algorithm	Cost	Status
ABC	0.5407	OPTIMUM ✓
GA	0.5407	OPTIMUM ✓
PSO	0.6129	Local Min ⚠



Additional Comparisons for Convergence Times

ABC, PSO and also Genetic Algorithm were used to compare the convergence times/iterations.



Our Another Original Contribution: Transition to the Attack Model

- Model Transition: Defense (interception) → Attack scenario.
- Addition of Decoys: 5 decoy targets were added alongside real missiles.
- Saturation: Decoys engage the air defense system, increasing the penetration probability of missiles.
- '0' State: The option of not assigning to a target was added.

Penetration Probability Formula

$$P_{pen} = P_{min} + (P_{max} - P_{min}) \times (1 - e^{(-\lambda \cdot N_j)})$$

Symbol	Description
N_j	number of decoys going to j-th target
λ (lambda)	Saturation coefficient = 0.5
P_{min}	Penetration without Decoy (base)
P_{max}	Penetration at full saturation

Added Operational Constraints

1

Budget Constraint

The total cost of missiles and decoys cannot exceed the operational budget ($B=100$).

2

Compatibility Constraint

If $C_{i,j} = 0$, the i -th missile cannot physically be assigned to the j -th target.

3

Decoy Logic Constraint

Assigning a decoy to a target without assigning a real missile is a waste and is penalized.

4

Salvo Firing Constraint

The Tgo difference of missiles heading to the same target must be less than ϵ .

Penalty Type	Coefficient	Weight
Q_j (Fixed Grouping)	0.1	Light
S_i (Multiple Assignment)	0.04	Light
U_j / L_j (Adaptive)	0.1	Light
w_{heavy} (Decoy, Compatibility)	100	Heavy
w_{light} (Salvo, Budget)	10	Medium

Mathematical Model

OBJECTIVE FUNCTION (To be Minimized)

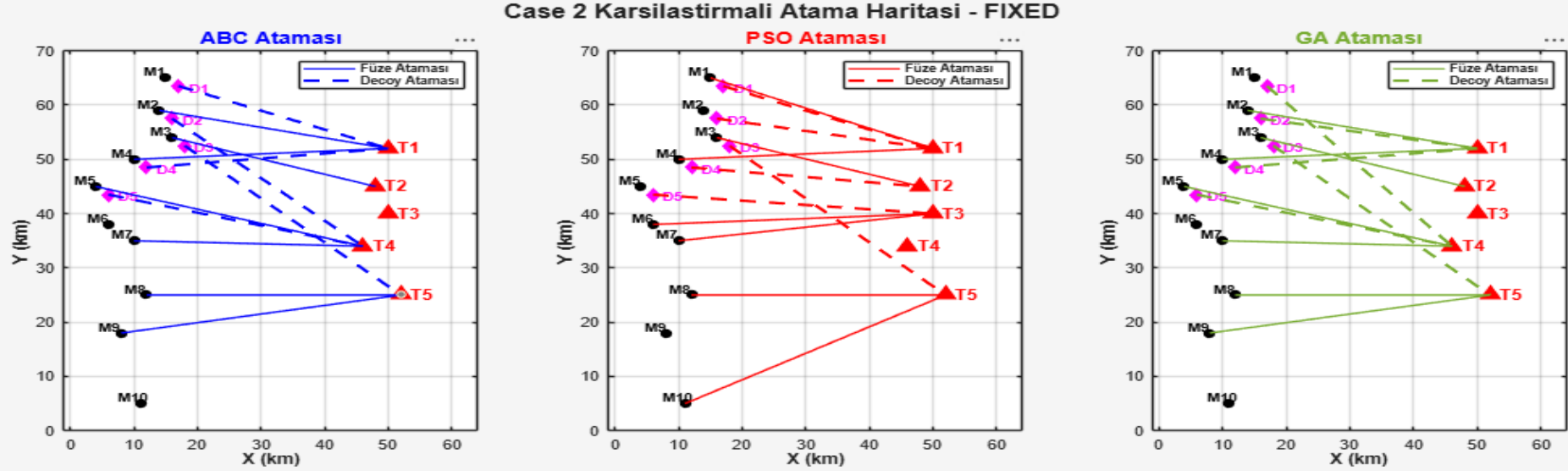
$$\min J = \sum V_j \times \prod (1 - P_{\text{succ}}(i,j) \times X_{i,j}) + \text{Penalty}$$

Individual Success Probability

$$P_{\text{succ}}(i,j) = P_{\text{pen}}(j) \times P_{\text{hit}}(i,j) \times P_{\text{kill}}(i,j)$$

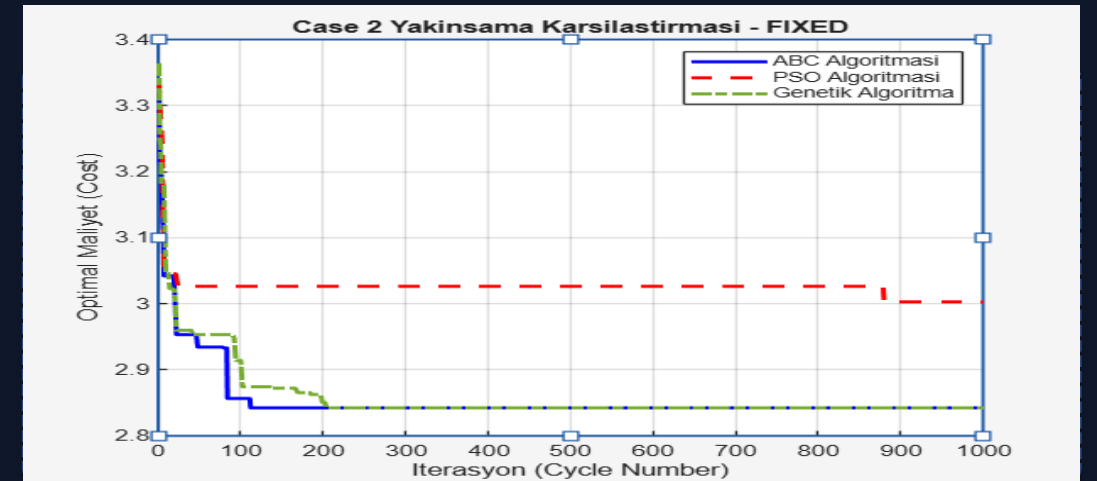
Symbol	Description	Example Value
V_j	importance/threat value of j-th target	0.85, 0.72, 0.80
$X_{i,j}$	assignment of i-th missile to j-th target (0/1)	Optimized by algorithm
$P_{\text{hit}}(i,j)$	Kinematic hit probability	Between 0.0 - 1.0
$P_{\text{kill}}(i,j)$	Kill probability on hit	0.85 (fixed)
$P_{\text{pen}}(j)$	Penetration probability	Between Pmin and Pmax (Formule given on the previous page)

Numerical Results (Attack Model) — Case 2 (5 Targets) - Fixed

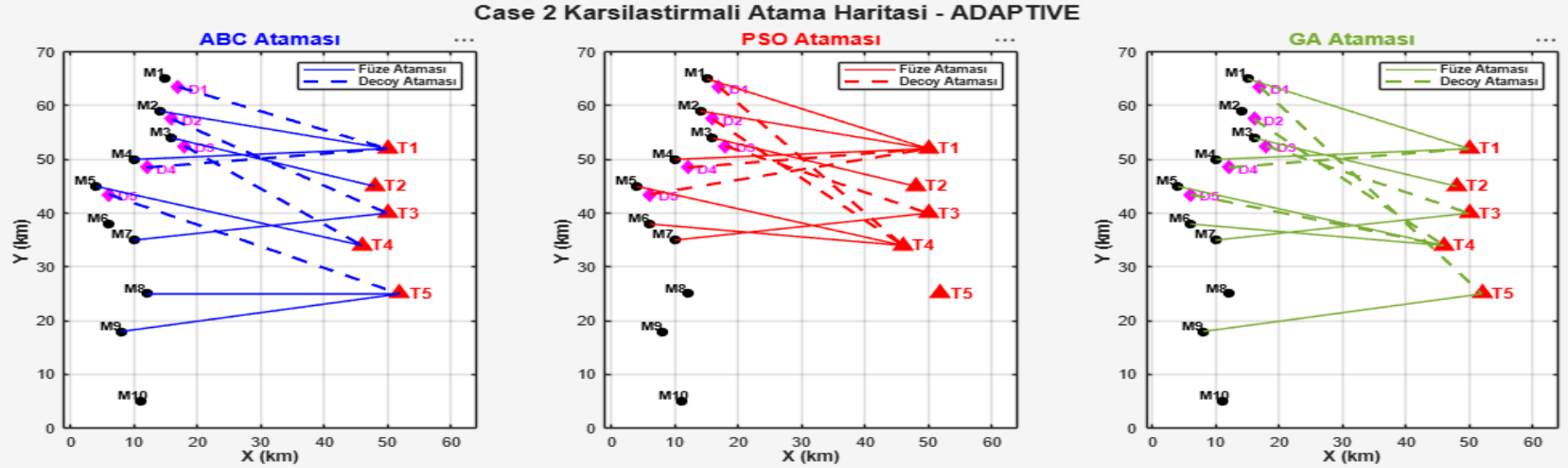


FIXED GROUPING

Algorithm	Cost	Status
ABC	2.8422	OPTIMUM ✓
GA	2.8422	OPTIMUM ✓
PSO	3.0027	Local Min ⚠

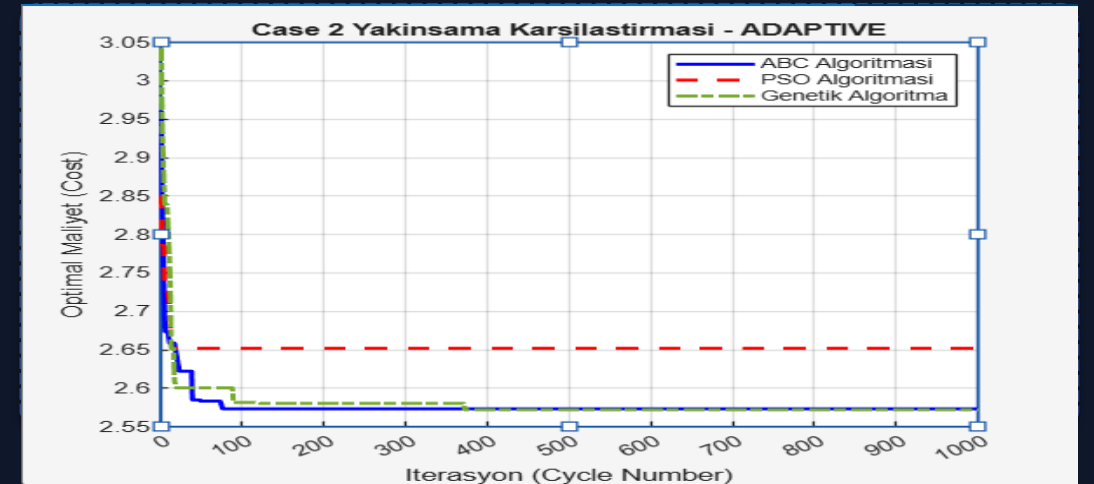


Numerical Results (Attack Model) — Case 2 (5 Targets) - Adaptive



ADAPTIVE GROUPING

Algorithm	Cost	Status
ABC	2.5720	OPTIMUM ✓
GA	2.5720	OPTIMUM ✓
PSO	2.7226	Local Min ⚠



Summary of All Results and Performance

Attack Scenario	Grouping	ABC Cost	GA Cost	PSO Cost	PSO Difference (%)
Case 1 (3 Targets)	Fixed	1.1610	1.1610	1.3535	+15.7%
Case 1 (3 Targets)	Adaptive	0.8610	0.8610	0.9213	+7.0%
Case 2 (5 Targets)	Fixed	2.8422	2.8422	3.0027	+5.64%
Case 2 (5 Targets)	Adaptive	2.5720	2.5720	2.7226	+5.85%

✓ Superiority of GA and ABC

GA's Crossover and Mutation operators enabled escaping from the local minimum traps that PSO fell into. Also, ABC uses an avoidance mechanism for local minimums. They found the unique optimum in both Defense and Attack scenarios.

Future Works

Steps to evolve the system into more realistic warfare environments:

3-Axis (3D) Engagement Model

- The current model considers targets in a 2D (x-y) plane.
- Including the altitude (z) axis of missiles and targets in the equation.
- Modeling how Pitch and Yaw angles affect the radar and Line of Sight.
- Taking terrain and radar horizon into account.

Dynamic (Moving) Target Engagement

- Modeling scenarios where targets do not follow fixed routes but continuously maneuver.
- In case the target performs an evasive maneuver after assignment, the algorithm re-assigns during flight (real-time).
- For this, reducing the algorithm's execution time to millisecond levels (GPU parallelization, etc.).

Thank You

Questions?

PROJECT SUMMARY

The defense model in the paper was verified; then expanded to a complex attack optimization model including decoys, budget constraints, and Genetic Algorithm support. GA and ABC outperformed PSO in all scenarios.

0.86

BEST ATTACK COST

3

ALGORITHM

6

CONSTRAINT TYPE

15

MATLAB FILE